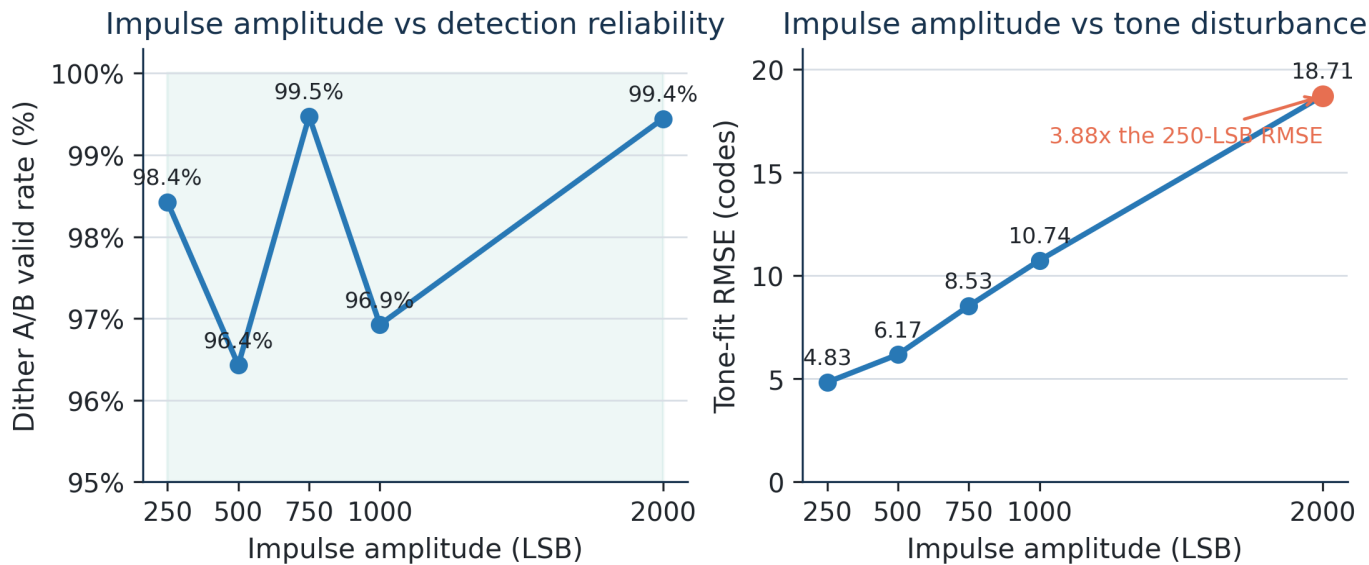


TIADC Impulse Dither - Board-Level Operating-Region Update

Research update for Professor Liu | 27 August 2026

Key takeaway. Board sweeps show that sparse, low-amplitude impulse excitation remains reliably detectable with limited disturbance over a practical operating region. Very narrow pulses are viable; excessively sparse repetition reduces detection. This strengthens the practical injection operating-region case, while the known gain and fine-skew limitations remain unchanged.

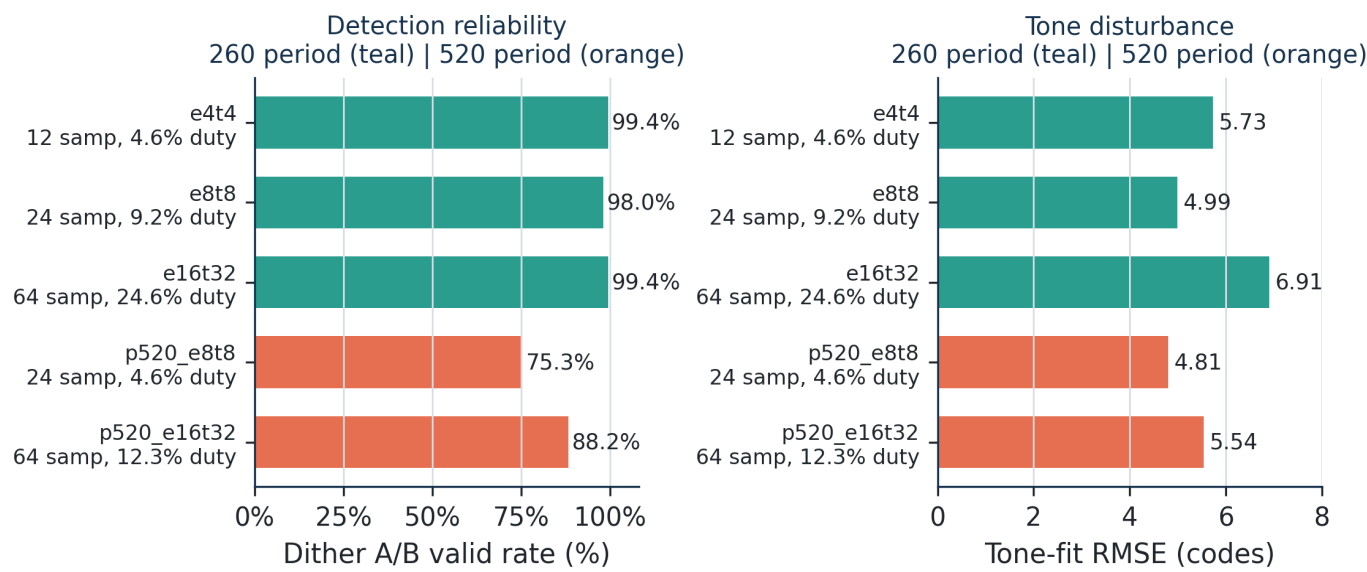
1. Board-Level Amplitude Sweep



Across 250-2000 LSB, A/B event validity is 96.4-99.5% and timing correlation is 0.99881-0.99993. Detection therefore remains strong across the tested range. The cost of stronger injection is visible in the tone fit: RMSE rises from 4.83 codes at 250 LSB to 18.71 codes at 2000 LSB (+13.89 codes; 3.88x). The sweep identifies a broad low-disturbance operating region, not a single globally optimal amplitude; larger impulses are not inherently better.

Figure 1. Values are means from the committed board timing/skew captures. The shaded detection band marks 95-100%, not a pass threshold.

2. Board-Level Pulse / Duty Sweep



At the standard 260-DAC-sample period, a 12-sample pulse occupies only 4.6% of the waveform and still gives 99.4% validity. The 24- and 64-sample cases give 98.0% and 99.4%. Doubling the period to 520 samples reduces validity to 75.3% (24-sample pulse) and 88.2% (64-sample pulse), despite similar tone RMSE. Impulses can be narrow and low-duty, but repetition should not be made arbitrarily sparse.

Figure 2. Duty is derived as pulse length / repetition period. The p520_e16t32 run has no calibration_performance.csv; no final-skew or Stage 5 value is assigned.

3. Implication for Selling Point 1

Evidence supports	Not yet demonstrated
Sparse impulse generation; reliable recovery through the existing DAC-to-ADC chain; low temporal occupancy; a measured strength/disturbance trade-off.	Independent passive coupling into an otherwise unmodified mission-signal path. A later coupling demo remains a publication-level option.

4. Status of Selling Point 2

Event detection is strong. Offset observability remains limited; dither gain has no stable calibration factor; fine-skew is unreliable under measured dispersion; bandwidth calibration was not pursued. Tone-based calibration remains the stable main path. No three-parameter separation is claimed.

5. Decisions for the September Meeting

1. Are the current board results sufficient to begin consolidating the work toward publication?
2. Should the next hardware step focus on an independent passive-coupling demonstration to strengthen selling point 1?
3. What scope and division of work should be assigned to the additional FPGA student?

Data note. All ten summary rows were independently recomputed from committed board CSV captures. No new board tests were run.